

Interfaces between PT and DT

Last edited by [Jonas Hohn](#) 5 months ago

TODO: For each machine:

- ☐ Add a short description of the inputs/outputs
- ☐ Suggest a mqtt topic name for each

Overall

- We want access to the raw data as used by the control software
- Sometimes, we want preprocessed data for ease of use

Components

Encoder Moter

- Inputs and outputs as given by the Fischertechnik spec
 - In
 - forward: boolean
 - reverse: boolean
 - Out
 - encoder1Level: boolean // High/low signal
 - encoder2Level: boolean
- Preprocessed
 - Relative steps measured by encoder, from encoder1Level & encoder2Level

Stations list

- Vacuum Gripper(AHe)

Machine port desc	Machine port number	Direction	Type	MQTT topic
reference switch vertical axis	I1	Out	Boolean	/\${machine-id}/ref-switch-vertical
reference switch horizontal axis	I2	Out	Boolean	/\${machine-id}/ref-switch-horizontal
reference switch rotate	I3	Out	Boolean	/\${machine-id}/ref-switch-rotation
encoder vertical axis impulse 1	B1	Out	Boolean	/\${machine-id}/motor-vertical-enc-1
encoder vertical axis impulse 2	B2	Out	Boolean	/\${machine-id}/motor-vertical-enc-2
encoder horizontal axis impulse 1	B3	Out	Boolean	/\${machine-id}/motor-horizontal-enc-1
encoder horizontal axis impulse 2	B4	Out	Boolean	/\${machine-id}/motor-horizontal-enc-2
encoder rotate impulse 1	B5	Out	Boolean	/\${machine-id}/motor-rotation-enc-1
encoder rotate impulse 2	B6	Out	Boolean	/\${machine-id}/motor-rotation-enc-2
Vertical position	Virtual	Out	Integer	/\${machine-id}/motor-vertical-pos
Horizontal position	Virtual	Out	Integer	/\${machine-id}/motor-horizontal-pos
Rotation position	Virtual	Out	Integer	/\${machine-id}/motor-rotation-pos
motor vertical axis up	Q1	In	Boolean	/\${machine-id}/move-vertical-up
motor vertical axis down	Q2	In	Boolean	/\${machine-id}/move-vertical-down
motor horizontal axis backward	Q3	In	Boolean	/\${machine-id}/move-horizontal-up
motor horizontal axis forward	Q4	In	Boolean	/\${machine-id}/move-horizontal-down
motor rotate clockwise	Q5	In	Boolean	/\${machine-id}/move-rotation-clockwise
motor rotate counterclockwise	Q6	In	Boolean	/\${machine-id}/move-rotation-counterclockwise
compressor	Q7	In	Boolean	/\${machine-id}/enable-compressor
Valve vacuum	Q8	In	Boolean	/\${machine-id}/enable-valve

- Multiprocessing Station(AHe)

Machine port desc	Machine port number	Direction	Type	MQTT Topic
reference switch turn-table (position vacuum)	I1	Out	Integer	/\${machine-id}/ref-switch-rotation-at-vacuum
reference switch turn-table (position belt)	I2	Out	Integer	/\${machine-id}/ref-switch-rotation-at-belt
light-barrier end of conveyor belt	I3	Out	Boolean	/\${machine-id}/light-barrier-conveyor-end
reference switch turn-table (position saw)	I4	Out	Integer	/\${machine-id}/ref-switch-turn-table-at-saw
reference switch vacuum (position turn-table)	I5	Out	Boolean	/\${machine-id}/ref-switch-vacuum-at-turn-table
reference switch oven feeder inside	I6	Out	Boolean	/\${machine-id}/ref-switch-feeder-inside
reference switch oven feeder outside	I7	Out	Boolean	/\${machine-id}/ref-switch-feeder-outside
reference switch vacuum (position oven)	I8	Out	Integer	/\${machine-id}/ref-switch-vacuum-at-oven
light-barrier oven	I9	Out	Boolean	/\${machine-id}/light-barrier-oven
motor turn-table clockwise	Q1 (M1)	In	Boolean	/\${machine-id}/move-turn-table-clockwise
motor turn-table counterclockwise	Q2 (M1)	In	Boolean	/\${machine-id}/move-turn-table-counterclockwise
motor conveyor belt forward	Q3 (M2)	In	Boolean	/\${machine-id}/move-converyor-forward
motor saw	Q4 (M3)	In	Boolean	/\${machine-id}/enable-saw
motor oven feeder retract	Q5 (M4)	In	Boolean	/\${machine-id}/move-oven-feeder-retract
motor oven feeder extend	Q6 (M4)	In	Boolean	/\${machine-id}/move-oven-feeder-extend
motor vacuum towards oven	Q7 (M5)	In	Boolean	/\${machine-id}/move-vacuum-to-oven
motor vacuum towards turn-table	Q8 (M5)	In	Boolean	/\${machine-id}/move-vacuum-to-turn-table
light oven	Q9	In	Boolean	/\${machine-id}/enable-oven-light
compressor	Q10	In	Boolean	/\${machine-id}/enable-compressor
valve vacuum	Q11	In	Boolean	/\${machine-id}/enable-valve-vacuum
valve lowering	Q12	In	Boolean	/\${machine-id}/enable-valve-move-vacuum
valve oven door	Q13	In	Boolean	/\${machine-id}/enable-valve-move-oven-door
valve feeder	Q14	In	Boolean	/\${machine-id}/enable-valve-move-feeder

- 3D-Gripper(AHe)

Machine port desc	Machine port number	Direction	Type	MQTT topic
reference switch claw	I1	Out	Boolean	/\${machine-id}/ref-switch-claw
pulse counter gripper(TODO: weird name?)	I2	Out	Integer	/\${machine-id}/enc-counter-claw
reference switch grip arm	I3	Out	Boolean	/\${machine-id}/ref-switch-gripper
pulse counter grip arm(TODO: weird name?)	I4	Out	Integer	/\${machine-id}/enc-counter-gripper
reference switch vertical axis	I5	Out	Boolean	/\${machine-id}/ref-switch-vertical
reference switch turntable	I6	Out	Boolean	/\${machine-id}/ref-switch-rotation
encoder vertical axis impuls 1	B1	Out	Boolean	/\${machine-id}/motor-vertical-enc-1

Machine port desc	Machine port number	Direction	Type	MQTT topic
encoder vertical axis impuls 2	B2	Out	Boolean	/\${machine-id}/motor-vertical-enc-2
encoder turntable impuls 1	B3	Out	Boolean	/\${machine-id}/motor-rotation-enc-1
encoder turntable impuls 2	B4	Out	Boolean	/\${machine-id}/motor-rotation-enc-2
position vertical	Virtual	Out	Integer	/\${machine-id}/motor-vertical-pos
position rotation	Virtual	Out	Integer	/\${machine-id}/motor-rotation-pos
motor gripper open	Q1 (M1)	In	Boolean	/\${machine-id}/open-gripper
motor gripper close	Q2 (M1)	In	Boolean	/\${machine-id}/close-gripper
motor grip arm forward	Q3 (M2)	In	Boolean	/\${machine-id}/move-horizontal-forward
motor grip arm back	Q4 (M2)	In	Boolean	/\${machine-id}/move-horizontal-backward
motor vertical axis down	Q5 (M3)	In	Boolean	/\${machine-id}/move-vertical-up
motor vertical axis up	Q6 (M3)	In	Boolean	/\${machine-id}/move-vertical-down
motor turntable right	Q7 (M4)	In	Boolean	/\${machine-id}/move-rotation-clockwise
motor turntable left	Q8 (M4)	In	Boolean	/\${machine-id}/move-rotation-counterclockwise

- Conveyor Belt

Machine port desc	Machine port number	Direction	Type	MQTT topic
phototransistor feed station	I1	Out	Boolean	/\${machine-id}/phototransistor-feed-station
phototransistor swap station	I2	Out	Boolean	/\${machine-id}/phototransistor-swap-station
pulse button	I3	Out	Boolean	/\${machine-id}/pulse-button
motor conveyor belt forward	Q1	In	Boolean	/\${machine-id}/move-conveyor-forward
motor conveyor belt backward	Q2	In	Boolean	/\${machine-id}/move-conveyor-backward

- High Bay

Machine port desc	Machine port number	Direction	Type	MQTT topic
reference switch horizontal axis	I1	Out	Boolean	/\${machine-id}/ref-switch-horizontal
light-barrier inside	I2	Out	Boolean	/\${machine-id}/light-barrier-inside
light-barrier outside	I3	Out	Boolean	/\${machine-id}/light-barrier-outside
reference switch vertical axis	I4	Out	Boolean	/\${machine-id}/ref-switch-vertical
trail sensor (signal 1, lower)	A1	Out	Boolean	/\${machine-id}/trail-sensor-lower
trail sensor (signal 2, upper)	A2	Out	Boolean	/\${machine-id}/trail-sensor-upper
Encoder horizontal Impuls 1	B1	Out	Boolean	/\${machine-id}/horizontal-enc-1
Encoder horizontal Impuls 2	B2	Out	Boolean	/\${machine-id}/horizontal-enc-2
Encoder vertical Impuls 1	B3	Out	Boolean	/\${machine-id}/vertical-enc-1
Encoder vertical Impuls 2	B4	Out	Boolean	/\${machine-id}/vertical-enc-2
reference switch cantilever front	I5	Out	Boolean	/\${machine-id}/ref-switch-cantilever-front
reference switch cantilever back	I6	Out	Boolean	/\${machine-id}/ref-switch-cantilever-back
position vertical	Virtual	Out	Integer	/\${machine-id}/vertical-pos
position horizontal	Virtual	Out	Integer	/\${machine-id}/horizontal-pos
motor conveyor belt forward	Q1	In	Boolean	/\${machine-id}/move-conveyor-forward
motor conveyor belt backward	Q2	In	Boolean	/\${machine-id}/move-conveyor-backward
motor horizontal towards rack	Q3	In	Boolean	/\${machine-id}/move-arm-to-rack
motor horizontal towards conveyor belt	Q4	In	Boolean	/\${machine-id}/move-arm-to-conveyor
motor vertical axis downward	Q5	In	Boolean	/\${machine-id}/move-vertical-down
motor vertical axis upward	Q6	In	Boolean	/\${machine-id}/move-vertical-up
motor cantilever forward	Q7	In	Boolean	/\${machine-id}/move-cantilever-forward
motor cantilever backward	Q8	In	Boolean	/\${machine-id}/move-cantilever-backward

- Punching Machine

Machine port desc	Machine port number	Direction	Type	MQTT topic
phototransistor goods in/out	I1	Out	Boolean	/\${machine-id}/phototransistor-goods-in-out
phototransistor punching machine	I2	Out	Boolean	/\${machine-id}/phototransistor-punching-machine
switch punching machine up	I3	Out	Boolean	/\${machine-id}/switch-punching-machine-up
switch punching machine down	I4	Out	Boolean	/\${machine-id}/switch-punching-machine-down
motor conveyor belt forward	Q1	In	Boolean	/\${machine-id}/move-conveyor-forward
motor conveyor belt backward	Q2	In	Boolean	/\${machine-id}/move-conveyor-backward
motor punching machine up	Q3	In	Boolean	/\${machine-id}/move-punching-machine-up
motor punching machine down	Q4	In	Boolean	/\${machine-id}/move-punching-machine-down

- Sorting Line

Machine port desc	Machine port number	Direction	Type	MQTT topic
pulse counter	I1	Out	Integer	/\${machine-id}/pulse-counter
light-barrier inlet	I2	Out	Boolean	/\${machine-id}/light-barrier-inlet
light-barrier behind color sensor	I3	Out	Boolean	/\${machine-id}/light-barrier-behind-color-sensor
color sensor	A4	Out	Integer	/\${machine-id}/color-sensor
light-barrier white	I5	Out	Boolean	/\${machine-id}/light-barrier-white
light-barrier red	I6	Out	Boolean	/\${machine-id}/light-barrier-red
light-barrier blue	I7	Out	Boolean	/\${machine-id}/light-barrier-blue
motor conveyor belt	Q1	In	Boolean	/\${machine-id}/move-conveyor
compressor	Q2	In	Boolean	/\${machine-id}/enable-compressor
valve first ejector	Q3	In	Boolean	/\${machine-id}/enable-valve-first-ejector
valve second ejector	Q4	In	Boolean	/\${machine-id}/enable-valve-second-ejector
valve third ejector	Q5	In	Boolean	/\${machine-id}/enable-valve-third-ejector

- Indexed Line

Machine port desc	Machine port number	Direction	Type	MQTT topic
-------------------	---------------------	-----------	------	------------

Machine port desc	Machine port number	Direction	Type	MQTT topic
push-button slider 1 front	I1	Out	Boolean	/\${machine-id}/button-slider-1-front
push-button slider 1 rear	I2	Out	Boolean	/\${machine-id}/button-slider-1-rear
push-button slider 2 front	I3	Out	Boolean	/\${machine-id}/button-slider-2-front
push-button slider 2 rear	I4	Out	Boolean	/\${machine-id}/button-slider-2-rear
phototransistor slider 1	I5	Out	Boolean	/\${machine-id}/phototransistor-slider-1
phototransistor milling machine	I6	Out	Boolean	/\${machine-id}/phototransistor-milling-machine
phototransistor loading station	I7	Out	Boolean	/\${machine-id}/phototransistor-loading-station
phototransistor drilling machine	I8	Out	Boolean	/\${machine-id}/phototransistor-drilling-machine
phototransistor conveyor belt swap	I9	Out	Boolean	/\${machine-id}/phototransistor-conveyor-swap
motor slider 1 backward	Q1	In	Boolean	/\${machine-id}/move-slider-1-backward
motor slider 1 forward	Q2	In	Boolean	/\${machine-id}/move-slider-1-forward
motor slider 2 backward	Q3	In	Boolean	/\${machine-id}/move-slider-2-backward
motor slider 2 forward	Q4	In	Boolean	/\${machine-id}/move-slider-2-forward
motor conveyor belt feed	Q5	In	Boolean	/\${machine-id}/move-conveyor-feed
motor conveyor belt milling machine	Q6	In	Boolean	/\${machine-id}/move-conveyor-milling-machine
motor milling machine	Q7	In	Boolean	/\${machine-id}/move-milling-machine
motor conveyor belt drilling machine	Q8	In	Boolean	/\${machine-id}/move-conveyor-drilling-machine
motor drilling machine	Q9	In	Boolean	/\${machine-id}/move-drilling-machine
motor conveyor belt swap	Q10	In	Boolean	/\${machine-id}/move-conveyor-swap

Comments